

CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the May/June 2014 series

9702 PHYSICS

9702/41

Paper 4 (A2 Structured Questions), maximum raw mark 100

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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Section A

- 1 (a) work done bringing unit mass from infinity (to the point) M1 A1 [2]
- (b) $E_p = -m\phi$ B1 [1]
- (c) $\phi \propto 1/x$ C1
- either at $6R$ from centre, potential is $(6.3 \times 10^7)/6$ ($= 1.05 \times 10^7 \text{ J kg}^{-1}$)
and at $5R$ from centre, potential is $(6.3 \times 10^7)/5$ ($= 1.26 \times 10^7 \text{ J kg}^{-1}$) C1
change in energy $= (1.26 - 1.05) \times 10^7 \times 1.3$ C1
 $= 2.7 \times 10^6 \text{ J}$ A1
- or change in potential $= (1/5 - 1/6) \times (6.3 \times 10^7)$ (C1)
change in energy $= (1/5 - 1/6) \times (6.3 \times 10^7) \times 1.3$ (C1)
 $= 2.7 \times 10^6 \text{ J}$ (A1) [4]
- 2 (a) the number of atoms in 12 g of carbon-12 M1 A1 [2]
- (b) (i) amount $= 3.2/40$
 $= 0.080 \text{ mol}$ A1 [1]
- (ii) $pV = nRT$
 $p \times 210 \times 10^{-6} = 0.080 \times 8.31 \times 310$ C1
 $p = 9.8 \times 10^5 \text{ Pa}$ A1 [2]
(do not credit if T in $^{\circ}\text{C}$ not K)
- (iii) either $pV = 1/3 \times Nm \langle c^2 \rangle$
 $N = 0.080 \times 6.02 \times 10^{23}$ ($= 4.82 \times 10^{22}$)
and $m = 40 \times 1.66 \times 10^{-27}$ ($= 6.64 \times 10^{-26}$) C1
 $9.8 \times 10^5 \times 210 \times 10^{-6} = 1/3 \times 4.82 \times 10^{22} \times 6.64 \times 10^{-26} \times \langle c^2 \rangle$ C1
 $\langle c^2 \rangle = 1.93 \times 10^5$
 $c_{\text{RMS}} = 440 \text{ m s}^{-1}$ A1 [3]
- or $Nm = 3.2 \times 10^{-3}$ (C1)
 $9.8 \times 10^5 \times 210 \times 10^{-6} = 1/3 \times 3.2 \times 10^{-3} \times \langle c^2 \rangle$ (C1)
 $\langle c^2 \rangle = 1.93 \times 10^5$
 $c_{\text{RMS}} = 440 \text{ m s}^{-1}$ (A1)
- or $1/2 m \langle c^2 \rangle = 3/2 kT$ (C1)
 $1/2 \times 40 \times 1.66 \times 10^{-27} \langle c^2 \rangle = 3/2 \times 1.38 \times 10^{-23} \times 310$ (C1)
 $\langle c^2 \rangle = 1.93 \times 10^5$
 $c_{\text{RMS}} = 440 \text{ m s}^{-1}$ (A1)
- (if T in $^{\circ}\text{C}$ not K award max 1/3, unless already penalised in (b)(ii))

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- 3 (a) *either* change in volume = $(1.69 - 1.00 \times 10^{-3})$
or liquid volume $\ll$ volume of vapour
work done = $1.01 \times 10^5 \times 1.69 = 1.71 \times 10^5$ (J) M1
A1 [2]
- (b) (i) 1. heating of system/thermal energy supplied to the system B1 [1]
2. work done on the system B1 [1]
- (ii) $\Delta U = (2.26 \times 10^6) - (1.71 \times 10^5)$ C1
 $= 2.09 \times 10^6$ J (3 s.f. needed) A1 [2]
- 4 (a) kinetic (energy)/KE/ E_k B1 [1]
- (b) *either* change in energy = 0.60 mJ
or max E proportional to (amplitude)²/equivalent numerical working B1
new amplitude is 1.3 cm B1
change in amplitude = 0.2 cm B1 [3]
- 5 (a) graph: straight line at constant potential = V_0 from $x = 0$ to $x = r$ B1
curve with decreasing gradient M1
passing through $(2r, 0.50V_0)$ and $(4r, 0.25V_0)$ A1 [3]
- (b) graph: straight line at $E = 0$ from $x = 0$ to $x = r$ B1
curve with decreasing gradient from (r, E_0) M1
passing through $(2r, \frac{1}{4}E_0)$ A1 [3]
(for 3rd mark line must be drawn to $x = 4r$ and must not touch x-axis)
- 6 (a) (i) energy = EQ C1
 $= 9.0 \times 22 \times 10^{-3}$
 $= 0.20$ J A1 [2]
- (ii) 1. $C = Q/V$
 $V = (22 \times 10^{-3})/(4700 \times 10^{-6})$ C1
 $= 4.7$ V A1 [2]
2. *either* $E = \frac{1}{2}CV^2$ C1
 $= \frac{1}{2} \times 4700 \times 10^{-6} \times 4.7^2$
 $= 5.1 \times 10^{-2}$ J A1 [2]
- or* $E = \frac{1}{2}QV$ (C1)
 $= \frac{1}{2} \times 22 \times 10^{-3} \times 4.7$
 $= 5.1 \times 10^{-2}$ J (A1)
- or* $E = \frac{1}{2}Q^2/C$ (C1)
 $= \frac{1}{2} \times (22 \times 10^{-3})^2/4700 \times 10^{-6}$
 $= 5.1 \times 10^{-2}$ J (A1)

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- (b) energy lost (as thermal energy) in resistance/wires/battery/resistor
(award only if answer in (a)(i) > answer in (a)(ii)2) B1 [1]
- 7 (a) graph: V_H increases from zero when current switched on
 V_H then non-zero constant
 V_H returns to zero when current switched off B1
B1
B1 [3]
- (b) (i) (induced) e.m.f. proportional to rate
of change of (magnetic) flux (linkage) M1
A1 [2]
- (ii) pulse as current is being switched on
zero e.m.f. when current in coil
pulse in opposite direction when switching off B1
B1
B1 [3]
- 8 (a) discrete and equal amounts (of charge)
allow: discrete amounts of $1.6 \times 10^{-19} \text{ C/elementary charge/e}$
integral multiples of $1.6 \times 10^{-19} \text{ C/elementary charge/e}$ B1 [1]
- (b) weight = qV/d
 $4.8 \times 10^{-14} = (q \times 680)/(7.0 \times 10^{-3})$
 $q = 4.9 \times 10^{-19} \text{ C}$ C1
A1 [2]
- (c) elementary charge = $1.6 \times 10^{-19} \text{ C}$ (allow $1.6 \times 10^{-19} \text{ C}$ to $1.7 \times 10^{-19} \text{ C}$)
either the values are (approximately) multiples of this
or it is a common factor
it is the highest common factor M0
C1
A1 [2]
- 9 (a) e.g. no time delay between illumination and emission
max. (kinetic) energy of electron dependent on frequency
max. (kinetic) energy of electron independent of intensity
rate of emission of electrons dependent on/proportional to intensity
(any three separate statements, one mark each, maximum 3) B3 [3]
- (b) (i) (photon) interaction with electron may be below surface
energy required to bring electron to surface B1
B1 [2]

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(ii) 1. threshold frequency = 5.8×10^{14} Hz A1 [1]

2. $\phi = hf_0$ C1
 $= 6.63 \times 10^{-34} \times 5.8 \times 10^{14}$
 $= 3.84 \times 10^{-19}$ (J) C1
 $= (3.84 \times 10^{-19}) / (1.6 \times 10^{-19})$
 $= 2.4$ eV A1 [3]

or

$hf = \phi + E_{\text{MAX}}$ (C1)
 chooses point on line and substitutes values E_{MAX} , f and h into
 equation with the units of the hf term converted from J to eV (C1)
 $\phi = 2.4$ eV (A1)

10 (a) energy required to separate the nucleons (in a nucleus)
 to infinity A1 [2]
 (allow reverse statement)

(b) (i) $\Delta m = (2 \times 1.00867) + 1.00728 - 3.01551$ C1
 $= 9.11 \times 10^{-3}$ u C1
 binding energy = $9.11 \times 10^{-3} \times 930$
 $= 8.47$ MeV A1 [3]
 (allow 930 to 934 MeV so answer could be in range 8.47 to 8.51 MeV)
 (allow 2 s.f.)

(ii) $\Delta m = 211.70394 - 209.93722$
 $= 1.76672$ u C1
 binding energy per nucleon = $(1.76672 \times 930) / 210$ C1
 $= 7.82$ MeV A1 [3]
 (allow 930 to 934 MeV so answer could be in range 7.82 to 7.86 MeV)
 (allow 2 s.f.)

(c) total binding energy of barium and krypton M1
 is greater than binding energy of uranium A1 [2]

Section B

11 (a) (i) inverting amplifier B1 [1]

(ii) gain is very large/infinite B1
 V^+ is earthed/zero B1
 for amplifier not to saturate, P must be (almost) earth/zero B1 [3]

(b) (i) $R_A = 100$ k Ω A1
 $R_B = 10$ k Ω A1
 $V_{\text{IN}} = 1000$ mV A1 [3]

(ii) variable range meter B1 [1]

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- 12 (a)** series of X-ray images (for one section/slice)
 taken from different angles
 to give image of the section/slice
 repeated for many slices
 to build up three-dimensional image (of whole object)
 M1
 M1
 A1
 M1
 A1 [5]
- (b)** deduction of background from readings
 division by three
 C1
 C1
 $P = 5 \quad Q = 9 \quad R = 7 \quad S = 13$
 (four correct 2/2, three correct 1/2)
 A2 [4]
- 13 (a)** e.g. noise can be eliminated/waveform can be regenerated
 extra bits of data can be added to check for errors
 cheaper/more reliable
 greater rate of transfer of data
 (1 each, max 2)
 B2 [2]
- (b)** receives bits all at one time
 transmits the bits one after another
 B1
 B1 [2]
- (c)** sampling frequency must be higher than/(at least) twice frequency to be sampled
either higher (range of) frequencies reproduced on the disc
or lower (range of) frequencies on phone
either higher quality (of sound) on disc
or high quality (of sound) not required for phone
 M1
 A1
 B1 [3]
- 14 (a)** reduction in power (allow intensity/amplitude)
 B1 [1]
- (b) (i)** attenuation = 2.4×30
 = 72 dB
 A1 [1]
- (ii)** gain/attenuation/dB = $10 \lg(P_2/P_1)$
 $72 = 10 \lg(P_{\text{IN}}/P_{\text{OUT}})$ or $-72 = 10 \lg(P_{\text{OUT}}/P_{\text{IN}})$
 ratio = 1.6×10^7
 C1
 C1
 A1 [3]
- (c)** e.g. enables smaller/more manageable numbers to be used
 e.g. gains in dB for series amplifiers are added, not multiplied
 B1 [1]